An Information-Theoretic View of Cache-Aided Networks: Part 1 – Coded Caching

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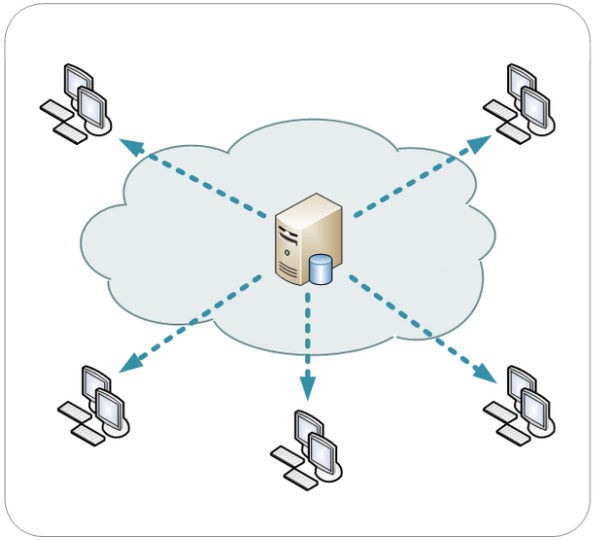
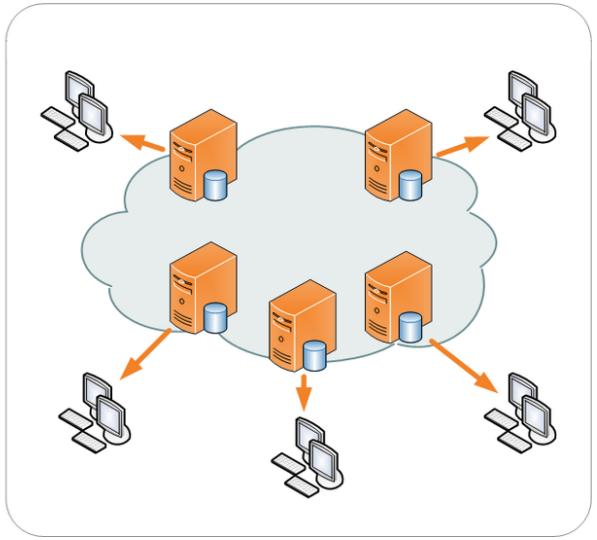
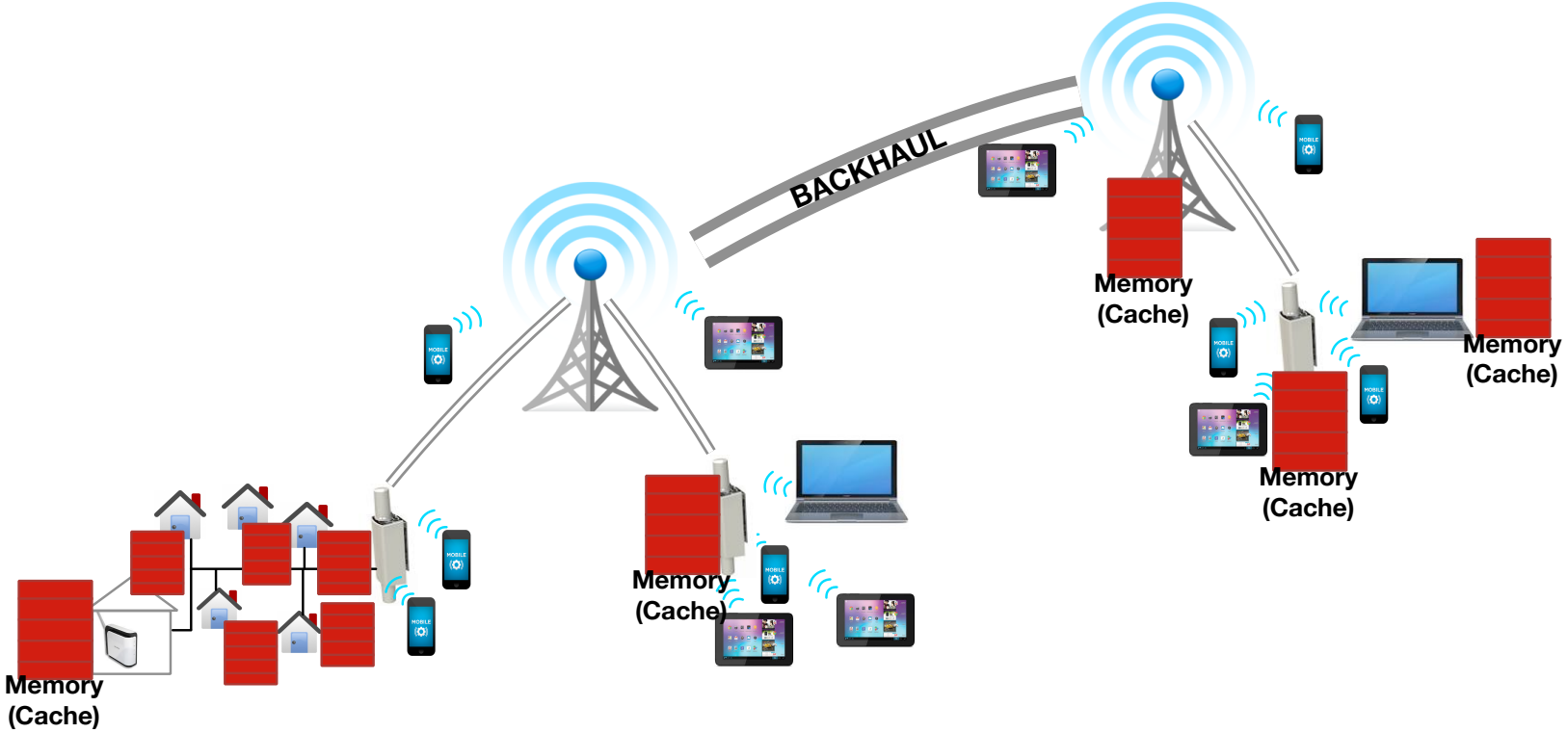
Content Delivery Networks





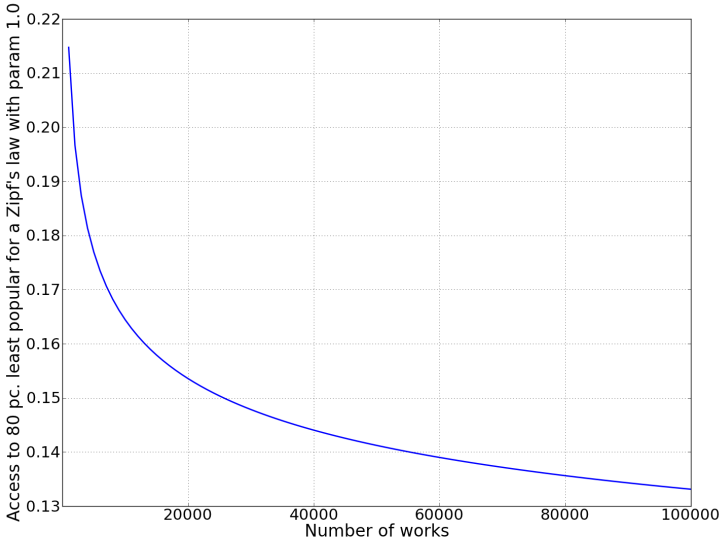
- Store contents in caches before file demands even known
- Reduce network load and latency during high-congestion periods
- Idea useful if certain files very popular and known in advance

Distributed Caches: Promising Solution for Cellular Networks



- Can cache at main BSs, picoBSs, femtoBSs, or directly at end users

File Popularities



- Static file popularity follows a Zipf distribution $P(x) = Cx^{-\alpha}$
- Evolution of file popularities (youtube videos) can also be predicted

Use pro-active caching to improve cellular systems!

- decrease network load
- decrease latency

Information-Theoretic View of Caching

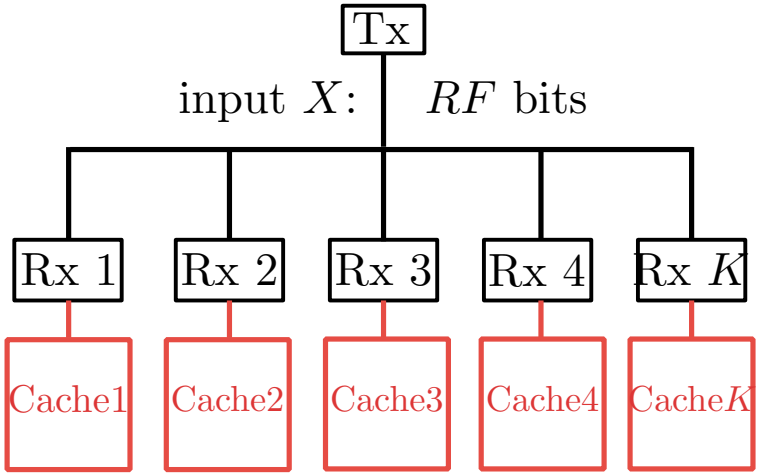
- Go beyond obvious *local* caching gain
- Create coding opportunities through smart cache placement
- Exploit multi-cast opportunities

  $\rightarrow$ serve many users/demands with same signals
- Global caching gain

  $\rightarrow$ receivers can profit from other receivers' cache memories

A Simple Network

<u>Library:</u> Files $W_1, W_2, \ldots, W_N$ of F bits each



- All files equally popular $\rightarrow$ we consider only most popular files
- All files equally large.
- Before the actual transmission there is an idle period where the transmitter (server) can fill the receivers' cache memories.
- Cache placement phase only constrained by memory sizes.

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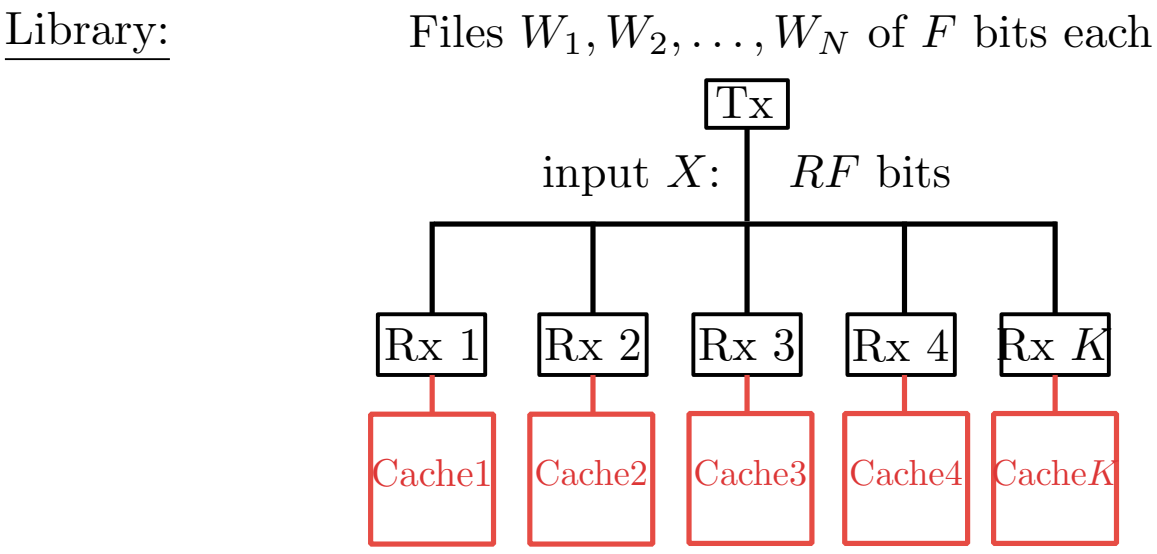
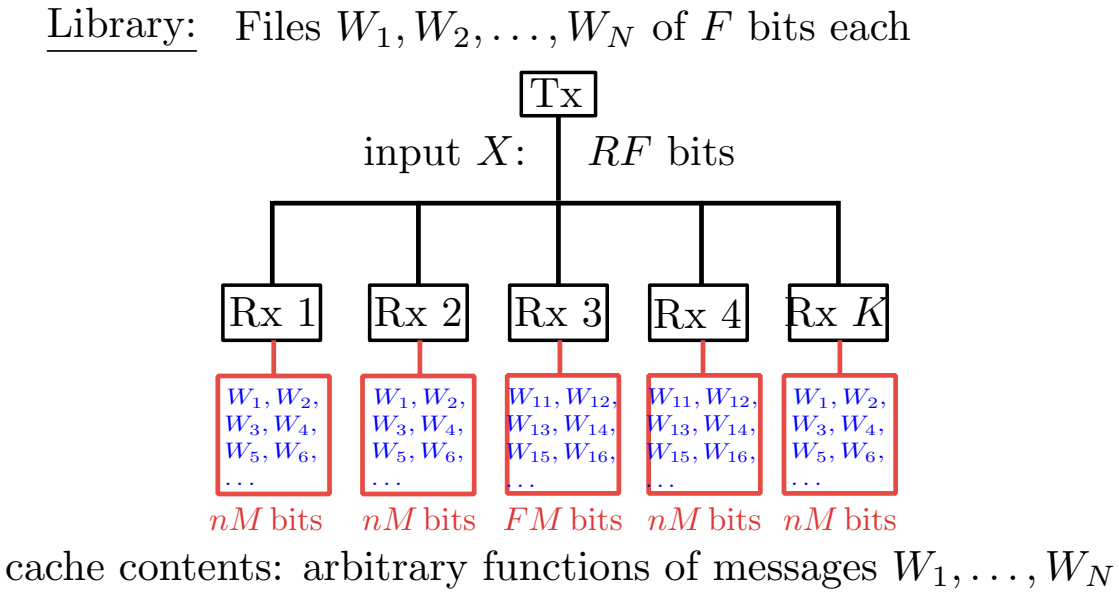
[1] M. A. Maddah-Ali, U. Niesen, "Fundamental Limits of Caching." *IEEE Transactions on Information Theory.*

A Simple Network



Communication in two phases:

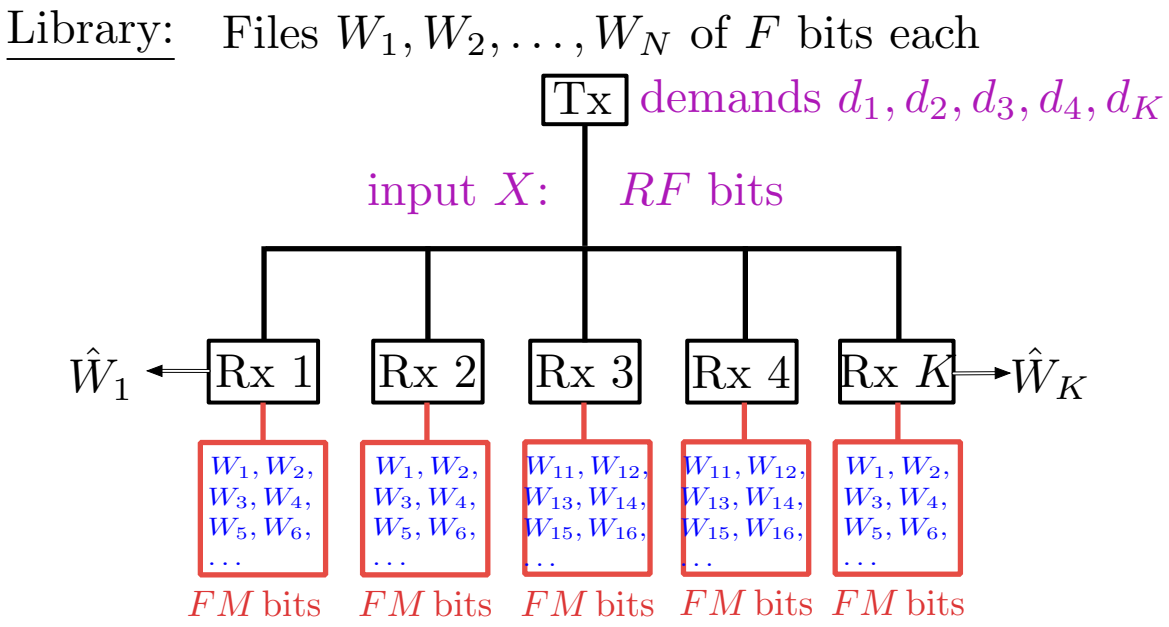
A Simple Network



Communication in two phases:

- Placement phase: Tx fills caches without knowing which receiver demands which message

A Simple Network



Communication in two phases:

- Placement phase: Tx fills caches without knowing which receiver demands which message
- Delivery phase:
  - Receiver k wants file W_{d_k} $\rightarrow$ sends demand d_k to transmitter
  - Tx describes $W_{d_1}, \ldots, W_{d_K}$ to Rxs $1, \ldots, K$ through input X
  - Tx describes demands $d_1, \ldots, d_K$ to all receivers

Fundamental Rate-Memory Tradeoff $R^\star(M)$

Definition (Achievable Rate-Memory Pairs)

A pair (R, M) is achievable if there exist placement and delivery schemes s.t. $\mathbb{P}\{\hat{W}_k \neq W_k\} \to 0$ as $F \to \infty$ for $k = 1, \dots, K$ and *all demands* $(d_1, \dots, d_K)$.

Definition (Fundamental Rate-Memory Tradeoff $R^\star(M)$)

$$R^\star(M) \quad := \quad \min\left\{R\colon \text{ such that } (R, M) \text{ achievable.}\right\}$$

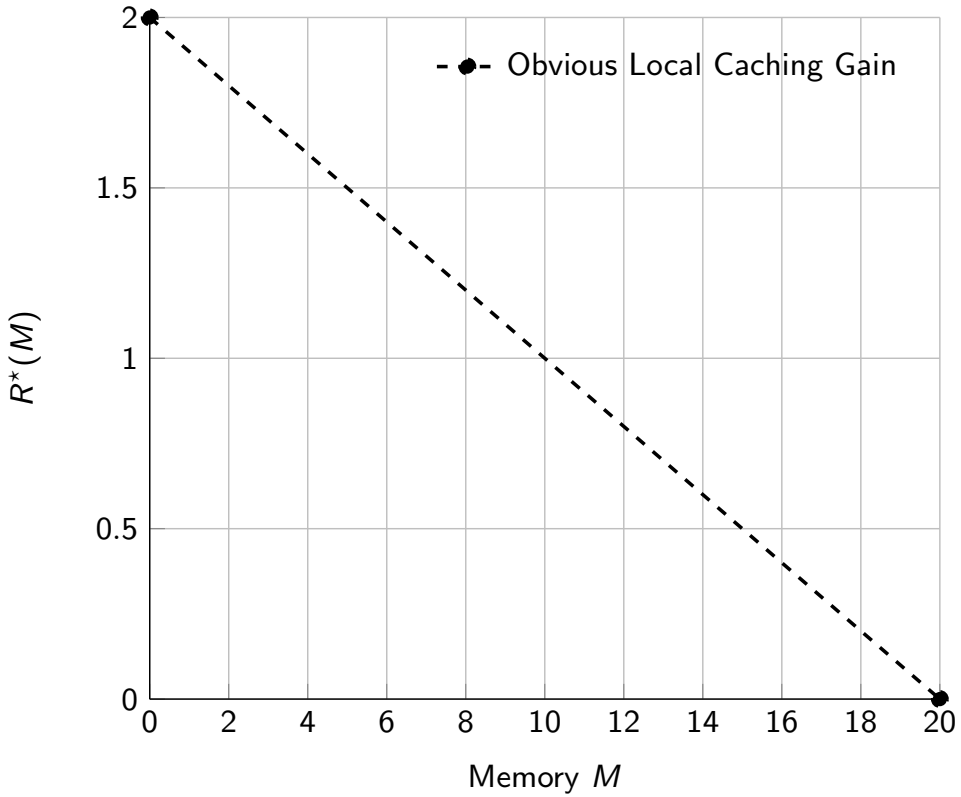
Some properties:

- $R^\star(M)$ is decreasing in M.
- $R^\star(M)$ is bounded above by $\min\{N, K\}$. Moreover:
$$R^\star(M = 0) = \min\{N, K\}.$$
- $R^\star(M)$ is nonnegative. Moreover:
$$R^\star(M) = 0, \qquad \forall M \geq N.$$

Obvious Upper Bound on $R^\star(M)$

$$R^\star(M) \leq \min\{K, N\}\left(1 - \frac{M}{N}\right).$$

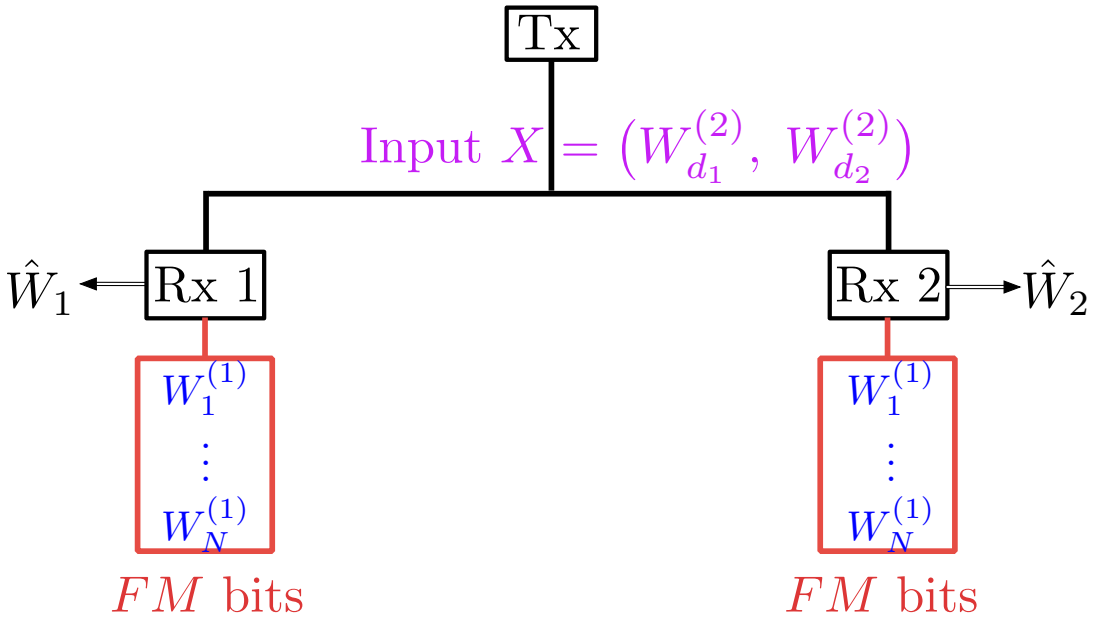
- Example with $N = 20$ files and $K = 2$ users



- Achieved through time/memory sharing or by the following naive scheme...

Naive Scheme for $K = 2$ Receivers

<u>Library:</u> Files $W_1, W_2, \ldots, W_N$ of F bits each



- Split W_d into two parts $(W_d^{(1)}, W_d^{(2)})$ of sizes $\frac{FM}{N}$ and $F\left(1 - \frac{M}{N}\right)$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ at both Rxs 1 and 2
- Delivery input $X = \left(W_{d_1}^{(2)},\ W_{d_2}^{(2)}\right)$ (if $d_1 \neq d_2$)

In the worst-case, delivery rate needs to be $R = 2\left(1 - \frac{M}{N}\right)$.

Naive Scheme for K Receivers

- Split W_d into two parts $(W_d^{(1)}, W_d^{(2)})$ of sizes $F\frac{M}{N}$ and $F\big(1-\frac{M}{N}\big)$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ *at all* rxs
- Deliver part $W_d^{(2)}$ for each demanded message W_d.
  - If $K \geq N$, in the worst case:

$$X = (W_{d_1}^{(2)},\ W_{d_2}^{(2)}, \ldots,\ W_{d_K}^{(2)}).$$

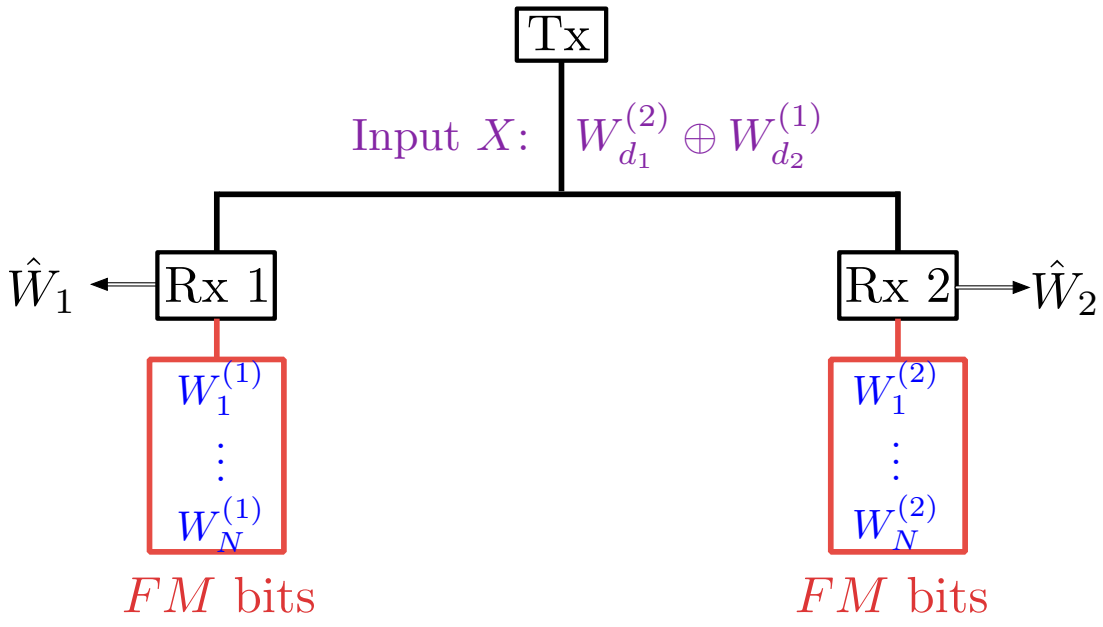
  - If $K < N$, in the worst case:

$$X = (W_1^{(2)},\ W_2^{(2)}, \ldots,\ W_N^{(2)}).$$

Required Delivery Rate is $\quad R = \min\{K, N\} \cdot \big(1 - \frac{M}{N}\big).$

Coded caching for $K = 2$ Receivers [Maddah-Ali&Niesen 2014]

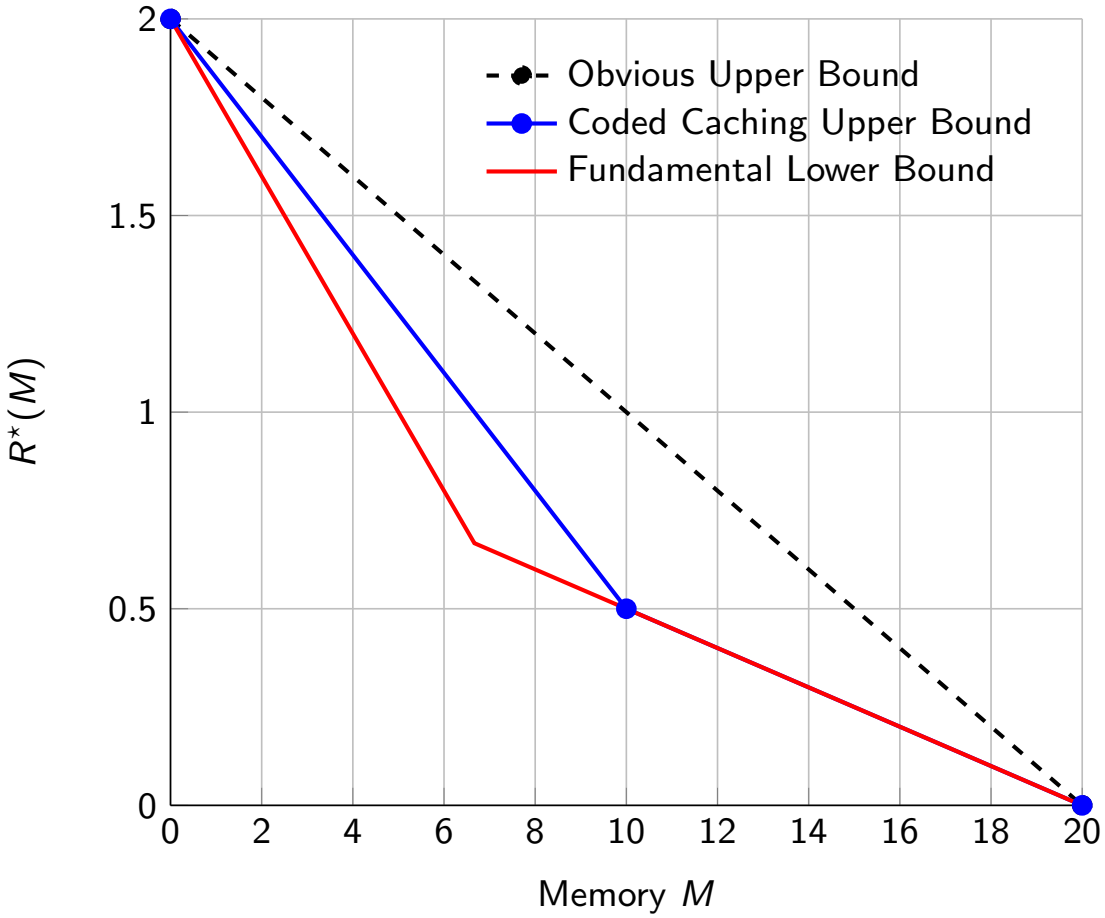
<u>Library:</u> Files $W_1, W_2, \ldots, W_N$ of F bits each



- Split W_d into two parts $(W_d^{(1)}, W_d^{(2)})$ each of $\frac{F}{2}$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ at Rx1 and part $W_d^{(2)}$ at Rx2
- Delivery input $X = W_{d_1}^{(2)} \oplus W_{d_2}^{(1)}$

Achieves Rate-Memory Pair $\quad M = \frac{N}{2} \quad$ and $\quad R = \frac{1}{2}$.

Fundamental Limit and Bounds for 2 Users ($N = 20$ files)



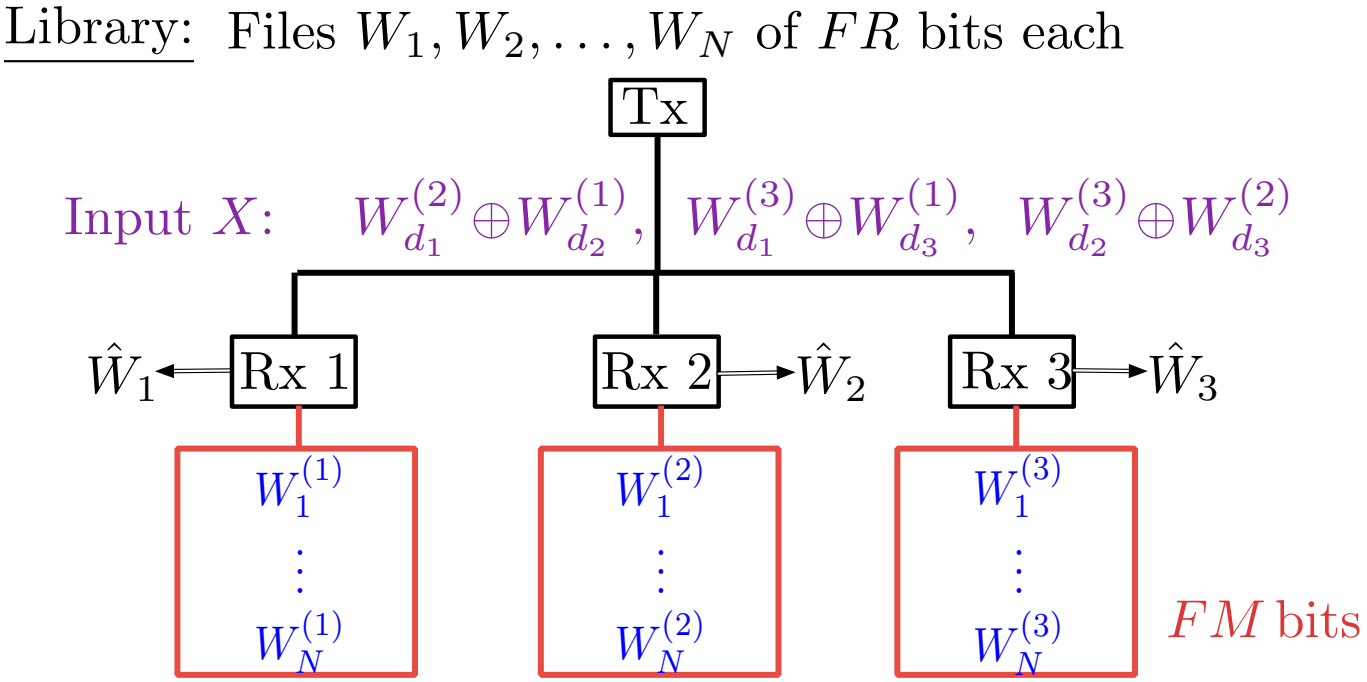
Time- and Memory Sharing for arbitrary parameter $\alpha \in [0, 1]$

- Assume Scheme 1 achieves (R_1, M_1) and Scheme 2 achieves (R_2, M_2)
- Split each file $W_d = (W_d^{(1)}, W_d^{(2)})$ consisting of αF and $(1-\alpha)F$ bits each.
- Apply placements of Scheme 1 to $\{W_d^{(1)}\}$ using $M_1 \alpha F$ bits of memory and placements of Scheme 2 to $\{W_d^{(2)}\}$ using $M_2(1-\alpha)F$ bits of memory
- Apply delivery of Scheme 1 to $\{W_d^{(1)}\} \to$ signal $X^{(1)}$ of $R_1 \alpha F$ bits; and apply delivery of Scheme 2 to $\{W_d^{(2)}\} \to$ signal $X^{(2)}$ of $R_2(1-\alpha)F$ bits

Total cache memory $MF = M_1 \alpha F + M_2(1-\alpha)F$; and
total number of delivery bits $RF = R_1 \alpha F + R_2(1-\alpha)F$

$$\Rightarrow (\alpha M_1 + (1-\alpha)M_2,\ \alpha R_1 + (1-\alpha)R_2) \quad \text{is achievable } \forall \alpha \in [0, 1]$$

Coded caching for $K = 3$ Receivers, Parameter $t = 1$

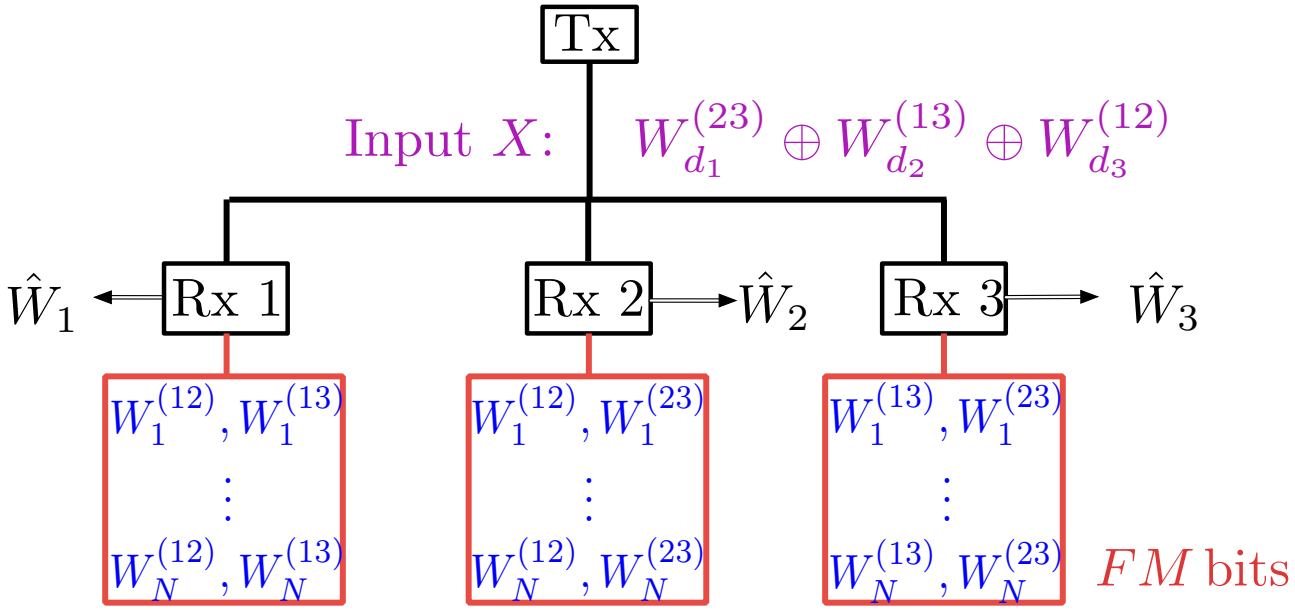


- Split W_d into three parts $(W_d^{(1)}, W_d^{(2)}, W_d^{(3)})$ each $\frac{F}{3}$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ at Rx1, part $W_d^{(2)}$ at Rx2, and part $W_d^{(3)}$ at Rx3
- Delivery input $X = W_{d_1}^{(2)} \oplus W_{d_2}^{(1)},\ W_{d_1}^{(3)} \oplus W_{d_3}^{(1)},\ W_{d_2}^{(3)} \oplus W_{d_3}^{(2)}$

Achieves Rate-Memory Pair $\quad M = \frac{N}{3} \quad$ and $\quad R = 1.$

Coded caching for $K = 3$ Receivers, Parameter $t = 2$

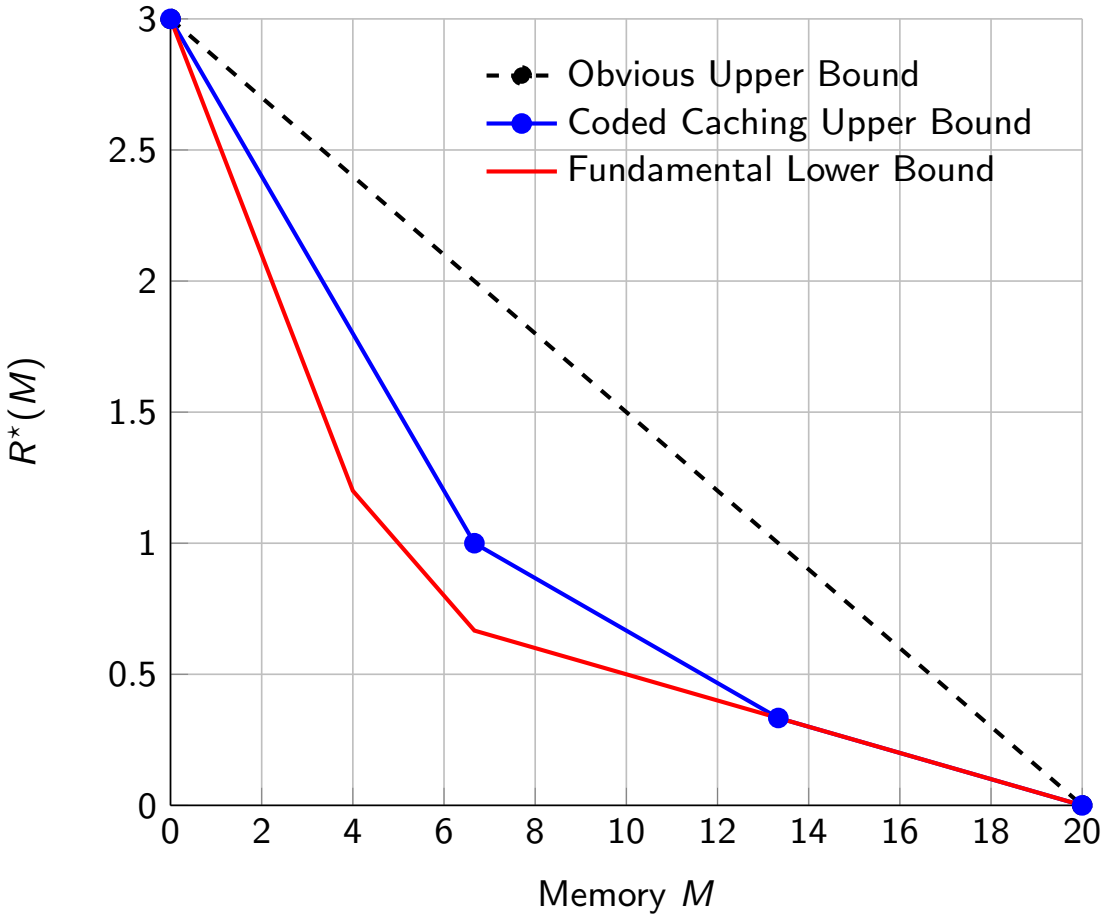
<u>Library:</u> Files $W_1, W_2, \ldots, W_N$ of F bits each



- Split W_d into three parts $(W_d^{(12)}, W_d^{(13)}, W_d^{(23)})$ each of $\frac{F}{3}$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(12)}$ at Rxs 1 and 2, part $W_d^{(13)}$ at Rxs 1 and 3, and part $W_d^{(23)}$ at Rxs 2 and 3
- Delivery input $X = W_{d_1}^{(23)} \oplus W_{d_2}^{(13)} \oplus W_{d_3}^{(12)}$

Achieves Rate-Memory Pair $\quad M = \frac{2N}{3} \quad$ and $\quad R = \frac{1}{3}$.

Bounds for 3 Users ($N = 20$ files)



Coded Caching for K Users

- Parameter $t \in \{1, \dots, K-1\}$

- *Placement:* Split each W_d into $\binom{K}{t}$ parts and save each part at a different subset of receivers
  Let for each size-t subset $\mathcal{G}$ denote $W_d^{\mathcal{G}}$ the part of W_d placed in caches of all receivers in $\mathcal{G}$.

- *Delivery transmission:* For each set $\mathcal{S} = \{s_1, \dots, s_{t+1}\}$ of size $t+1$, send

$$W_{\text{XOR},\mathcal{S}} := \bigoplus_{\ell=1}^{t+1} W_{d_{s_\ell}}^{(\mathcal{S}\setminus\{s_\ell\})}$$

- *Delivery reception:* Receiver s_j has stored in its cache memory

$$W_{d_{s_\ell}}^{(\mathcal{S}\setminus\{s_\ell\})}, \qquad \forall \ell \in \{1, \dots, j-1, j+1, \dots, t\}.$$

  So, with $W_{\text{XOR},\mathcal{S}}$ it can recover $W_{d_{s_j}}^{(\mathcal{S}\setminus\{s_j\})}$.

  This way it can recover all missing parts of $W_{d_{s_j}}$.

Analysis of Coded Caching for K Users

- Fix parameter $t \in \{1, \dots, K-1\}$
- Each part of a file is of size

$$F \cdot \binom{K}{t}^{-1} \quad \text{bits.}$$

- Each receiver stores $\binom{K-1}{t-1}$ parts of each file. So the placement requires cache memory

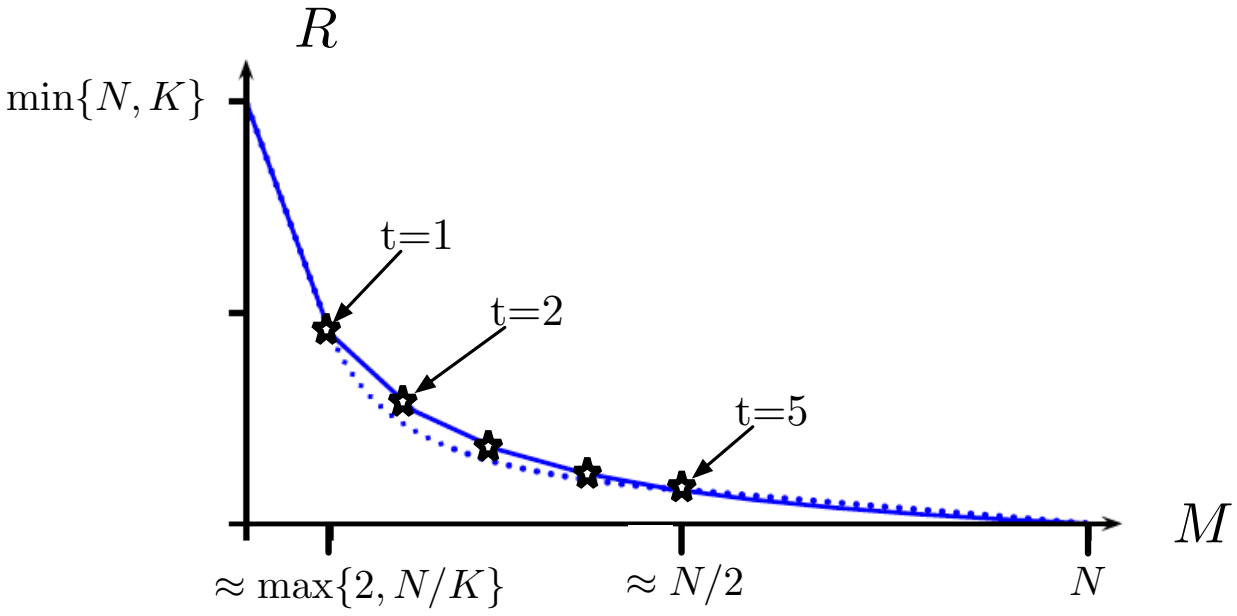
$$M = N \frac{\binom{K-1}{t-1}}{\binom{K}{t}} = N \frac{t}{K}. \qquad \text{(increasing in } t\text{)}$$

- Coded caching sends an XOR-message to each subset of $t+1$ receivers. So the total rate is

$$R = \frac{\binom{K}{t+1}}{\binom{K}{t}} = \frac{K-t}{t+1}. \qquad \text{(decreasing in } t\text{)}$$

Performance of Coded Caching

- $K = 6$



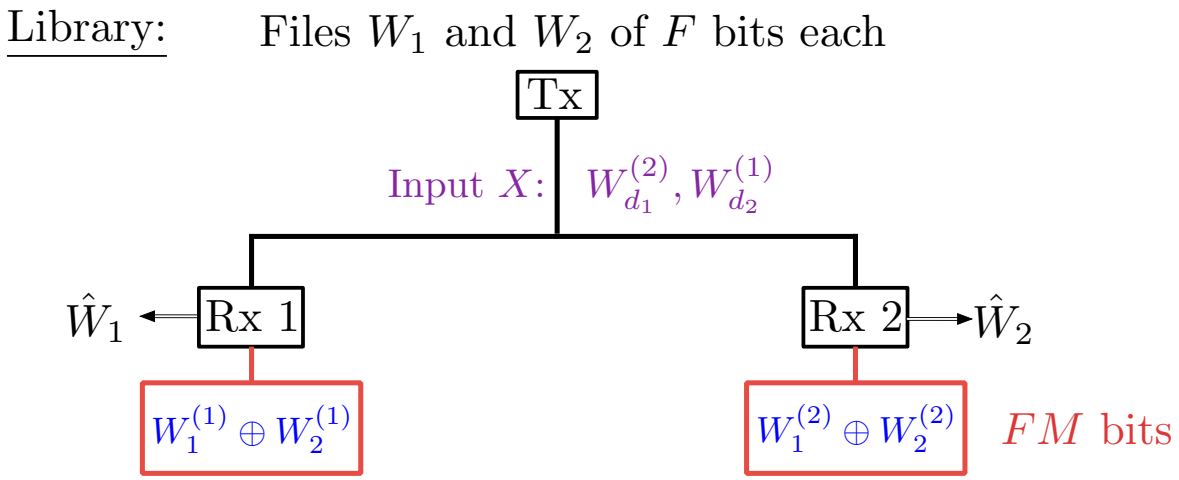
Coded Caching Upper Bound

For all $M \in \frac{N}{K} \cdot \{0, 1, \ldots, K-1, K\}$:

$$R^\star(M) \leq \min\left\{K\left(1 - \frac{M}{N}\right)\left(1 + \frac{MK}{N}\right)^{-1},\ N\left(1 - \frac{M}{N}\right)\right\}.$$

Achievability can be Improved!

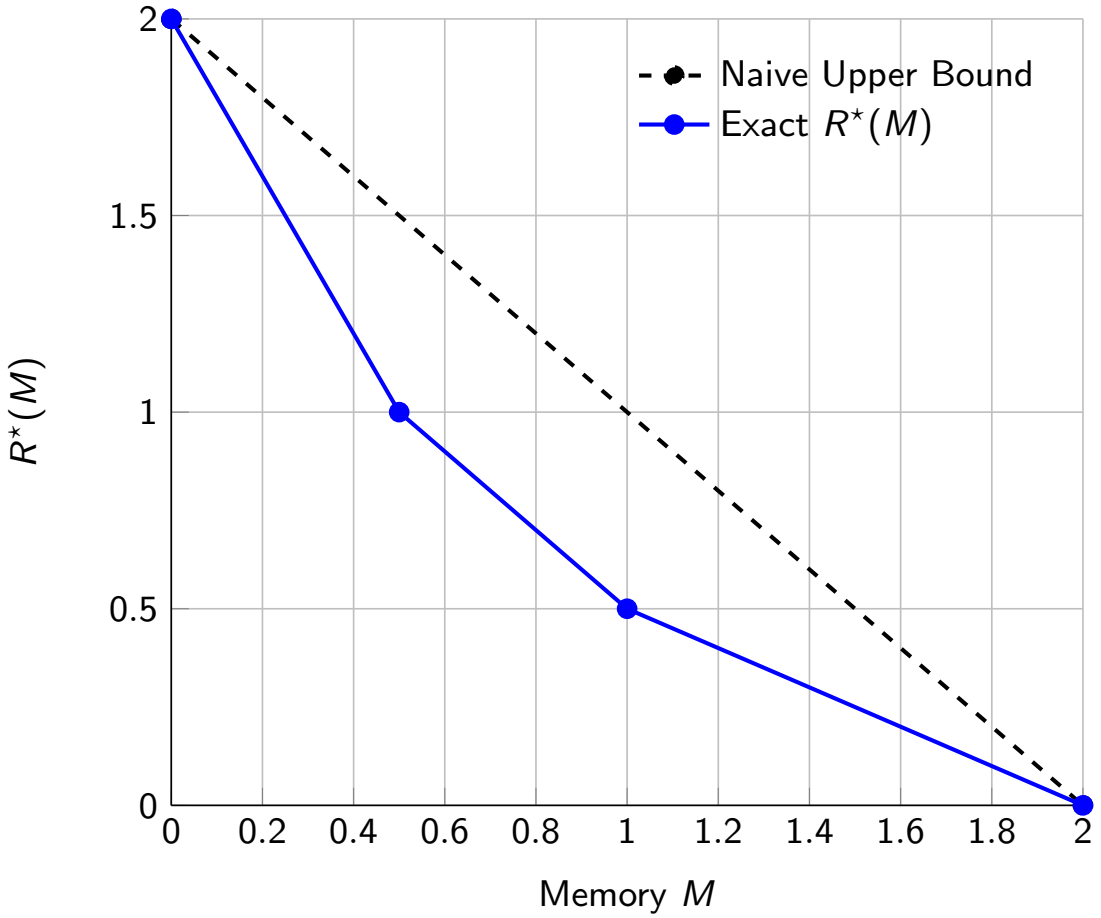
Example: $K = 2$ and $N = 2$:



- Split W_d into two parts $(W_d^{(1)}, W_d^{(2)})$ each of $\frac{F}{2}$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ at Rx1 and part $W_d^{(2)}$ at Rx2
- Delivery input $X = W_{d_1}^{(2)}, W_{d_2}^{(1)}$

Achieves Rate-Memory Pair $\quad M = \frac{1}{2} \quad$ and $\quad R = 1.$

Exact Rate-Memory Tradeoff $R^\star(M)$ for $K = N = 2$



First Lower Bound $R \geq s - \frac{s}{\lfloor N/s \rfloor} M$

- Consider only Receivers $1, \dots, s$, where $s \leq \min\{N, K\}$
- Consider demand vectors

$$\begin{aligned} \mathbf{d}^{(1)} &:= (1, \dots, s) \\ \mathbf{d}^{(2)} &:= (s+1, \dots, 2s) \\ &\dots \\ \mathbf{d}^{(\lfloor N/s \rfloor)} &:= ((\lfloor N/s \rfloor - 1)s + 1, \dots, \lfloor N/s \rfloor s) \end{aligned}$$

- Let $X^{(\ell)}$ denote the delivery input for demand $\mathbf{d}^{(\ell)}$
- From $X^{(1)}, \dots, X^{(\lfloor N/s \rfloor)}$ and $Z_1, \dots, Z_s$ one can calculate $W_1, \dots, W_{\lfloor N/s \rfloor s}$:

$$\begin{aligned} & H\big(X^{(1)}, \dots, X^{(\lfloor N/s \rfloor)}, Z_1, \dots, Z_s\big) \geq H(W_1, \dots, W_{\lfloor N/s \rfloor s}) \\ \iff \quad & H\big(X^{(1)}, \dots, X^{(\lfloor N/s \rfloor)}\big) + H\big(Z_1, \dots, Z_s\big) \geq H(W_1, \dots, W_{\lfloor N/s \rfloor s}) \\ \iff \quad & FR\lfloor N/s \rfloor + sFM \geq \lfloor N/s \rfloor sF \\ \iff \quad & R \geq s - \frac{s}{\lfloor N/s \rfloor} M. \end{aligned}$$

Second Lower Bound $R \geq s - \frac{s^2}{N} M$

- Consider only Receivers $1, \ldots, s$, where $s \leq \min\{N, K\}$
- $\mathcal{D}_s$: all demand vectors $\mathbf{d}_s$ of s users having s different demands.

$$|\mathcal{D}_s| = \binom{N}{s} s!$$

- Let $X^{(\mathbf{d}_s)}$ denote the delivery input for demand $\mathbf{d}_s$
- $\forall \mathbf{d}_s \in \mathcal{D}_s$ it holds that:

$$\begin{aligned} F \cdot R &\geq H(X^{\mathbf{d}_s}) \geq I(X^{\mathbf{d}_s}; W_{d_1}, \ldots, W_{d_s}, Z_1, \ldots, Z_s) \\ &\geq I(X^{\mathbf{d}_s}; W_{d_1}, \ldots, W_{d_s} | Z_1, \ldots, Z_s) \\ &= H(W_{d_1}, \ldots, W_{d_s} | Z_1, \ldots, Z_s). \end{aligned}$$

- By averaging over all demands $\mathbf{d}_s \in \mathcal{D}_s$ and by Han's inequality:

$$\begin{aligned} FR &\geq \frac{s}{N}\Big(H(W_1, \ldots, W_N) - I(W_1, \ldots, W_N; Z_1, \ldots, Z_s)\Big) \\ &\geq F\left(s - \frac{s^2}{N} M\right). \end{aligned}$$

Third Lower Bound $R \geq s - \sum_{i=1}^{s} \frac{i}{N-i+1} M$

- Consider only Receivers $1, \ldots, s$, where $s \leq \min\{N, K\}$
- $\mathcal{D}_s$: all demand vectors $\mathbf{d}_s$ of s users having s different demands.

$$|\mathcal{D}_s| = \binom{N}{s} s!$$

- Let $X^{(\mathbf{d}_s)}$ denote the delivery input for demand $\mathbf{d}_s$
- $\forall \mathbf{d}_s \in \mathcal{D}_s$ it holds that:

$$F \cdot R \quad \geq \quad sF - \sum_{i=1}^{s} I(W_{d_i}; Z_1, \ldots, Z_i | W_{d_1}, \ldots, W_{d_{i-1}})$$

- By averaging over all demands $\mathbf{d}_s \in \mathcal{D}_s$

$$\begin{aligned} FR \quad &\geq \quad sF - \sum_{i=1}^{s} \frac{1}{\binom{N}{s} s!} \sum_{\mathbf{d}_s \in \mathcal{D}_s} I(W_{d_i}; Z_1, \ldots, Z_i | W_{d_1}, \ldots, W_{d_{i-1}}) \\ &\geq \quad F \left(s - \sum_{i=1}^{s} \frac{i}{N-i+1} M \right). \end{aligned}$$

Gap between Upper and Lower Bounds

- Multiplicative Gap:

$$\frac{\text{Best Upper Bound}}{\text{Best Lower Bound}} \leq 2.315.$$

- Upper bound from a more constrained scenario

$$R_{\text{Dec}} := \frac{N-M}{M}\left(1-\left(1-\frac{M}{N}\right)^{\min\{K,N\}}\right)$$

- Third lower bound is piecewise linear over intervals $[M_{\ell+1}, M_\ell)$ with

$$M_\ell = \begin{cases} \frac{N-\ell}{\ell+1} & \text{if } \ell \in \{0,1,\ldots,\min\{K,N\}-1\}, \\ 0 & \text{if } \ell = \min\{N,K\}. \end{cases}$$

- Make R_{Dec} piecewise linear over same intervals.
- Ratio is bilinear and it suffices to consider end-points of intervals.
  $\rightarrow$ check all end-points!